

Project Title: NLP Assistant - PHP Chatbot and Text Analyzer with Supabase

Selected Stack:

- Frontend: HTML
- Backend: PHP
- Database: Supabase
- NLP Tools: NLTK and spaCy
- Supporting Assets: CSS, Bootstrap, JavaScript

Overview:

The system accepts typed text or browser voice input, sends it to PHP endpoints, calls a Python NLP processor using NLTK and spaCy, returns analysis and chatbot responses, and stores conversation records in Supabase.

Implemented NLP Concepts:

1. Tokenization - NLTK displays tokenized user input.
2. Sentiment Analysis - NLTK VADER classifies Positive, Negative, or Neutral.
3. Voice-to-Text - browser microphone converts speech to text.
4. Text-to-Speech - Speak button reads chatbot output aloud.
5. Chatbot System - user vs bot interface with stored Supabase history.
6. Keyword Extraction - spaCy-assisted important term extraction.
7. Text Classification - Complaint, Inquiry, Feedback, Command, or General.
8. Entity Extraction - spaCy entities when the English model is installed.

System Workflow:

User input -> JavaScript fetch -> PHP endpoint -> Python NLP processor -> Supabase save -> JSON response -> UI display.

Supabase Table: chats

Fields: id, user_message, bot_response, tokens, keywords, sentiment, sentiment_score, classification, entities, created_at.

How to Run:

1. Run database/nlp_assistant.sql in the Supabase SQL Editor.
2. Copy .env.example to .env and set SUPABASE_URL and SUPABASE_KEY.
3. python3 -m venv .venv && source .venv/bin/activate
4. pip install -r requirements.txt
5. python3 -m spacy download en_core_web_sm
6. php -S 127.0.0.1:8000
7. Open <http://127.0.0.1:8000>

Defense Demo Checklist:

Show chatbot, token list, sentiment result, keywords, classification, voice input, speech output, and Supabase history table.